CHAPTER 2

Complex numbers: the geometric point of view

Complex plane, modulus, argument, trigonometric form

Objectives. Represent a complex number by a point or a vector; determine a modulus and an argument; move from the algebraic form to the trigonometric form and back; interpret a modulus as a distance.

Prerequisites. Chapter 1; coordinates in the plane, distance between two points; unit circle, radians, standard values; scalar product and determinant of two vectors.

Fluency

Fluency · No calculator, five minutes.

1. Compute the distance between $A(1,-2)$ and $B(4,2)$.
2. Give $\cos\frac{3\pi}{4}$ and $\sin\frac{3\pi}{4}$.
3. Determine the real number $\theta \in (-\pi,\pi]$ such that $\cos\theta = -\frac{1}{2}$ and $\sin\theta = \frac{\sqrt{3}}{2}$.
4. Give the principal measure of the angle $\frac{17\pi}{6}$.
5. Are the vectors $\vec{u}\begin{pmatrix}2\\1\end{pmatrix}$ and $\vec{v}\begin{pmatrix}-1\\2\end{pmatrix}$ orthogonal?
6. Write $(2+i)(1+i)$ in algebraic form.